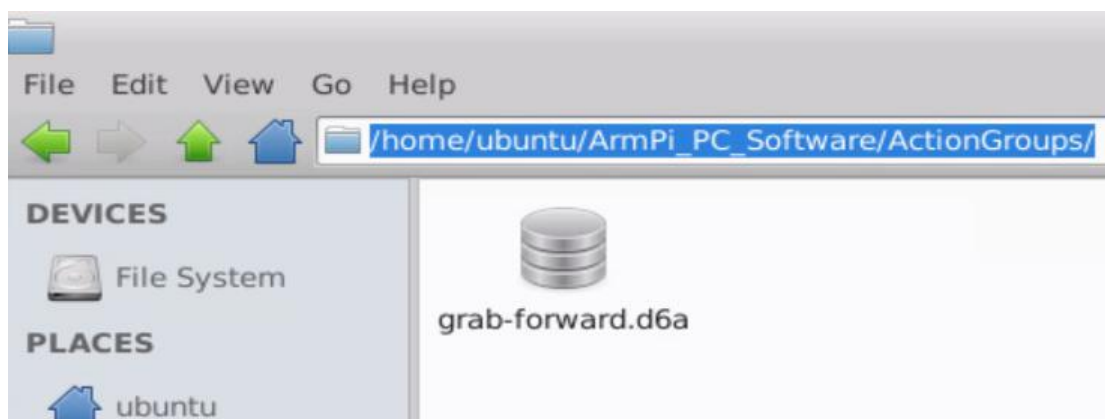


Lesson 3 Execute Action

1. Project Outcome

Call and execute the action group programmed in the “Lesson 2 Action Group Programming” through the terminal command line.

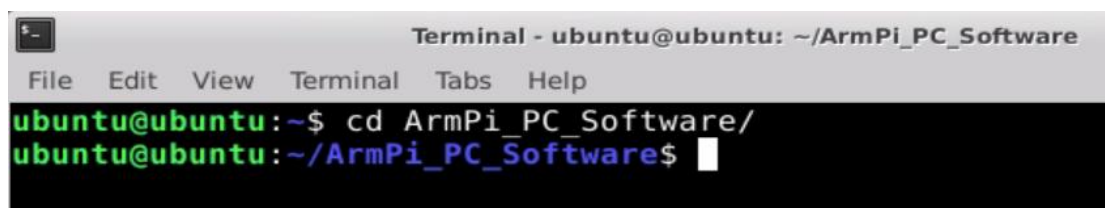
(Action group file must be saved in the path of **/home/ubuntu/ArmPi_PC_Software/ActionGroups/**)



2. Complete Project

2.1 Action Executes

1) Click “Applications” at the bottom left corner and enter the “**cd ArmPi_PC_Software/**” command to switch to PC software directory.



2) Enter “ls” command to check the files.

```
ubuntu@ubuntu:~/ArmPi_PC_Software$ ls
ActionGroupControl.py  ArmUi.py  ServoCmd.py  resource.qrc
ActionGroupControlDemo.py  ArmUi.ui  Ui.py.backup  resource_rc.py
ActionGroups  BusServoCmd.py  __pycache__
ArmPi.py  BusServoControl.py  images
ubuntu@ubuntu:~/ArmPi_PC_Software$
```

2.2 Single Action Execution

If you want to run the single action, you're required to modify in the file of actions programming.

- 1) Enter "**sudo vim ActionGroupControlDemo.py**" and press "Enter" to open the program through vi editor.
- 2) Press "i" key to delete "import Board" in the program.

```
1 import time
2 import Board
3 import ActionGroupControl as AGC
4
```

- 3) In the program, the default running actions are No.1 and No.2. And we need to change "1" and "2" to the name of execute action group. For example, press "i" key and replace "1" in parentheses with "grab-forward".

```
26 AGC.runAction('grab-forward') #
27 AGC.runAction('2')
-- INSERT --
```

Attention: the action name should match the file name, otherwise it will fail to call the action group.

- 4) We need to enter "#" in front of second line to execute single action.

```
26 AGC.runAction('grab-forward') #
27 #AGC.runAction('2')
-- INSERT --
```

5) Press “Ese” and enter “:” to enter last line command mode. Enter “wq” and press “Enter” to save and exit.

```
26 AGC.runAction('grab-forward')
27 #AGC.runAction('2')
:wq
```

6) Enter the command “**sudo python3 ActionGroupControlDemo.py**” again and press “Enter”. ArmPi will run “grab-forward” action once.

Note: if there is no action group file under the path, the LX terminal will display "Action Group File Cannot Be Found."

2.3 Execute More Actions

Refer to the “Lesson 2 Action Group Programming” to study how to program and save action. We will take the example of the third action programmed and saved.



1) Please enter “**sudo vim ActionGroupControlDemo.py**” command to the program file according to the above steps.

2) Copy the AGC.runAction(‘ ’) function. The name of the parameters in the function must be the same as the saved action file, otherwise it cannot be executed.

3) For example, after executing the No.1 and No.2 actions, call to run the No.3 action. Select the function in line 26 with the mouse, press the "Y key" twice, and then press "P" to copy the above function.

```
25 AGC.runAction('grab-forward') #
26 AGC.runAction('01.Hiwonder')
27 AGC.runAction('wave')
28
```

4) Revise the parameter in the function to “**wave**”, save and exit. Then enter the command “**sudo python3 ActionGroupControlDemo.py**” again.